More on Linear Systems and Matrices

6 October 2025

Study Guide

This week, we will be learning about:

- homogeneous linear systems, and their solutions;
- matrices and their properties;
- matrix equality, addition, and scalar multiplication;
- matrix multiplication;
- the transpose of a matrix.

Definitions

Definition 1. A linear system is said to be **homogeneous** if all the equations in the system are homogeneous. That is,

$$\left\{\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & 0 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & 0 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & 0 \end{array}\right. .$$

Every homogeneous linear system is consistent—that is, at least one solution must exist, given by the **trivial solution**

$$x_1 = x_2 = \cdots = x_n = 0.$$

If there are other solutions, they are called **nontrivial**.

Definition 2. A **matrix** is a rectangular array of numbers. The numbers in the array are called the **entries** in the matrix. The **size** of a matrix is described in terms of the number of rows and columns it contains.

Definition. A matrix with n rows and n columns is called a **square matrix of order n**. The entries $a_{11}, a_{22}, \ldots, a_{nn}$ are called the **diagonal entries** of the matrix. The entries a_{ij}, where $i \neq j$, are called the **non-diagonal entries** of the matrix. A square matrix whose non-diagonal entries are all zero is called a **diagonal matrix**. Notationally,

$$\boldsymbol{A} = \left(a_{ij}\right) \text{ is diagonal} \quad \Leftrightarrow \quad a_{ij} = 0 \text{ when } i \neq j.$$

Definition 3. A diagonal matrix is an **identity matrix** if all its diagonal entries are equal to 1. We let I_n denote the identity matrix of order n. A **zero matrix** is a matrix whose entries are all zero; we let $\mathbf{0}_{m\times n}$ denote the $m \times n$ zero matrix.

Definition 4. A square matrix in which all the entries above the diagonal are zero is called **lower triangular**, and a matrix in which all the entries below the diagonal are zero is called **upper triangular**. Notationally,

$$\boldsymbol{A} = \left(a_{ij}\right) \text{ is lower triangular} \quad \Leftrightarrow \quad a_{ij} = 0 \text{ when } i < j,$$

$$\boldsymbol{A} = \left(a_{ij}\right) \text{ is upper triangular} \quad \Leftrightarrow \quad a_{ij} = 0 \text{ when } i > j.$$

Definition 5. Two matrices are defined to be **equal** if they have the same size and their corresponding entries are equal.

$$\boldsymbol{A}_{m\times n} = \boldsymbol{B}_{m\times n} \quad \Leftrightarrow \quad a_{ij} = b_{ij} \text{ for all } i, j$$

If $\boldsymbol{A}$ and $\boldsymbol{B}$ are matrices of the same size, then the **sum** $\boldsymbol{A}+\boldsymbol{B}$ is the matrix obtained by adding the entries of $\boldsymbol{B}$ to the corresponding entries of $\boldsymbol{A}$.

$$\boldsymbol{A}+\boldsymbol{B} = \left(a_{ij}\right) + \left(b_{ij}\right) = \left(a_{ij}+b_{ij}\right).$$

If $\boldsymbol{A}$ is any matrix and c is any scalar, then the (scalar) **product** $c\boldsymbol{A}$ is the matrix obtained by multiplying each entry of the matrix $\boldsymbol{A}$ by c.

$$c\boldsymbol{A} = c\left(a_{ij}\right) = \left(c\cdot a_{ij}\right).$$

Note that the sum $\boldsymbol{A}+\boldsymbol{B}$ is a matrix that has the same size as $\boldsymbol{A}$ and $\boldsymbol{B}$. Similarly, the scalar product $c\boldsymbol{A}$ has the same size as $\boldsymbol{A}$.

Definition 6. Let $\boldsymbol{A}_{m\times p} = \left(a_{ij}\right)$ and $\boldsymbol{B}_{p\times n} = \left(b_{ij}\right)$ be matrices. The **matrix product** $\boldsymbol{AB}$ is the $m\times n$ matrix whose (i,j)-entry is computed by multiplying the corresponding entries of the ith row of $\boldsymbol{A}$ with the jth column of $\boldsymbol{B}$, and taking the sum of the resulting products. That is,

$$(\boldsymbol{AB})_{ij} = \sum_{k=1}^{p} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj}.$$

Definition 7. Let $\boldsymbol{A}$ be an $m\times n$ matrix. The **transpose of** $\boldsymbol{A}$, denoted by $\boldsymbol{A}^T$, is the $n\times m$ matrix that results by interchanging the rows and columns of $\boldsymbol{A}$. A square matrix $\boldsymbol{A}$ is said to be **symmetric** if $\boldsymbol{A} = \boldsymbol{A}^T$.

Exercises

1. For what values of a will the following linear system

$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + 5z = 2 \\ 4x + y + \left(a^2-14\right)z = a+2 \end{cases}$$

   have (a) no solution? (b) exactly one solution? (c) infinitely many solutions?

2. For what values of a and b will the following linear system

$$\begin{cases} ax + y = a \\ x + y + z = 1 \\ y + az = b \end{cases}$$

   have (a) no solution? (b) exactly one solution? (c) infinitely many solutions?

3. Find a formula for the quadric surface $ax^2+by^2+cz^2 = d$ that passes through the points $(1,1,1)$, $(1,3,3)$, and $(-2,0,2)$.

4. Prove the result on the transpose of the product matrix:

$$(\boldsymbol{AB})^T = \boldsymbol{B}^T\boldsymbol{A}^T.$$